

How to predict the zenith tropospheric delay (ZTD): using the ensemble forecast data from different NWP

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Abstract

Tropospheric delay is a major source of error affecting the accuracy of satellite earth observation, including global navigation satellite system (GNSS). The accurate zenith tropospheric delay (ZTD) prediction is significant for positioning and navigation. This study proposed a method for predicting ZTD using meteorological ensemble forecast data and analyzed their performances from different numerical weather prediction (NWP). Specifically, the ensemble forecast data from the European Centre for Medium-Range Weather Forecasts (ECMWF-IFS) and the U.S. National Centers for Environmental Prediction (NCEP-GFS) were utilized to predict the ZTD, and a total of 450 IGS ground stations worldwide were selected as reference data sources for evaluation. The comprehensive analysis of errors, spatial and temporal characteristics indicated that ensemble forecast data for ZTD prediction can achieve high accuracy, with the RMSE reaching 38.57 mm and 30.66 mm for ECMWF-IFS and NCEP-GFS, respectively. In terms of spatial distribution, the prediction accuracy of two types of ensemble forecast data varied with latitude and elevation. In terms of temporal characteristics, ensemble forecast data can provide reliable ZTD predictions across different initialization times, indicating that both types of data are capable of reasonably capturing the daily variations and regional characteristics of ZTD.

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1. Introduction

Tropospheric delay refers to the time delay caused by the refraction of electromagnetic signals as they pass through the neutral, non-ionized atmosphere below an altitude of 50 km (Bevis et al., 1992). This delay varies with the satellite's elevation angle and can range from 2 to 20 m, making it a significant source of error in various space

geodetic techniques (Yang et al., 2021a; Huang et al., 2023). Due to the difficulty and instability in directly measuring tropospheric delay, it is often modeled and estimated using the zenith tropospheric delay (ZTD), which represents the total delay in the vertical direction when the satellite elevation angle is 90°. ZTD can be further divided into two physical components, the zenith hydrostatic delay (ZHD), caused by the dry component of the atmosphere, and the zenith wet delay (ZWD), induced by the water vapor component (Yang et al., 2023a; Huang et al., 2024). Accurate estimation and modeling of ZTD is crucial for mitigating tropospheric delay and improving

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the accuracy of GNSS positioning and navigation (Yang et al., 2021b; Wang et al., 2025; Hao et al., 2025).

As a priori information in high-precision Global Navigation Satellite System (GNSS) data processing, empirical ZTD models are generally categorized into two types, the first of which is based on measured meteorological data (Zhu et al., 2025), as it requires temperature, pressure, and relative humidity for calculation, and includes classical models such as the Saastamoinen (Saastamoinen, 1972), Hopfield (Hopfield, 1969), and the Black model (Black, 1984). These models utilize atmospheric physical properties for ZTD estimation and require meteorological sensors at GNSS sites. However, due to the lack of meteorological sensors or insufficient coverage of meteorological stations, as well as the model construction errors, the accuracy of these models may be limited. Therefore, numerous researchers have proposed to improve the model to enhance the estimation accuracy of ZTD. Yang et al. (2021a) proposed two methods to improve the Hopfield and Saastamoinen model. One method was to add annual and semi-annual periodic terms, and the other method was based on Back-Propagation Artificial Neural Network (BP-ANN) to improve the model. Ding et al. (2016) used high-precision GNSS-ZTD data from the International GNSS Service (IGS) to analyze the residuals of the Saastamoinen model and then modeled them using a back propagation neural network. Based on this approach, a new real-time surface meteorology-based ZTD model, ISAAS, was proposed, which showed improved accuracy with bias and the RMSE errors lower than those from the Saastamoinen model in Russian test regions.

The second category of models does not rely on meteorological data but instead utilizes time series to construct models. For instance, He et al. (2024) developed a global ZTD forecasting model named LSTM based on long short-term memory neural networks using GNSS time series data. Chen et al. (2019) developed the SHATrop model using ZTD products from 223 stations of the Crustal Movement Observation Network of China (CMONOC) and further extended it to SHATrop-Extend by incorporating spatiotemporal variations and terrain height effects. Yuan et al. (2024) evaluated the SHATropE model in terms of geographic distribution, altitude, seasonal and diurnal variations using CMONOC data, and found that it performed better in high-altitude areas and at different UTC hours. Time series constructed from reanalysis data can also be used to estimate ZTD, as exemplified by models such as the UNB series, Global Zenith Tropospheric Delay (GZTD) series (Yao et al., 2013), IGGtrop (Li et al., 2018), and GPT series (Lagler et al., 2013; Böhm et al., 2015; Boehm et al., 2007). The UNB model, proposed by Collins and Langley, has evolved over time into versions like UNB1–4, UNB3m, and UNB.na (Leandro et al., 2008; Leandro et al., 2006; Leandro et al., 2009). The EGNOS model (Penna et al., 2001), used in the European EGNOS satellite-based augmentation system, provided meteorological parameters for ZTD computation with an accuracy

comparable to the Hopfield and Saastamoinen models using actual measurements. Li et al. (2012) developed the IGGtrop model based on reanalysis data from the National Centers for Environmental Prediction (NCEP). Li et al. (2018) further improved the model based on empirical vertical reduction functions and established two improved tropospheric delay models, IGGtrop_SH and IGGtrop_rH. The ZTD correction performance of the IGGtrop_SH model was significantly improved compared with that of the IGGtrop model, especially in the mid-latitude regions, while there was no significant change in the Southern Hemisphere. Zhang et al. (2016) evaluated the IGGtrop, EGNOS, and UNB3m models using GNSS-derived ZTD data from 25 CMONOC stations, concluding that these models showed significant seasonal variations, with IGGtrop performing best in China. Yao et al. (2013) used grid time series of global ZTD data from 2002 to 2009 provided by GGOS Atmosphere to study the global spatiotemporal variations of ZTD and developed the GZTD model based on spherical harmonics. Experimental results showed that the GZTD model outperformed models like UNB3m, UNB4, UNB3, and EGNOS, which only consider latitudinal ZTD variations. The GPT series models were constructed using reanalysis data from the European Centre for Medium-Range Weather Forecasts (ECMWF) and can provide direct ZTD correction. Yang et al. (2021b) analyzed the GPT3 model using 2018 ZTD products from the global IGS stations and found that GPT3 exhibited latitude-dependent bias, hemispherical differences, and minor seasonal variation. These models offered the advantage of direct ZTD estimation without the need for additional meteorological inputs, making them suitable for GNSS stations lacking weather sensors, though their accuracy may vary depending on geography and terrain.

Based on reanalysis time series, TU Wien (TUW) and the German Research Centre for Geosciences (GFZ) have respectively developed VMF3-based tropospheric delay products using ray-tracing techniques, namely TUW-VMF3 and GFZ-VMF3. Li et al. (2024, 2025) conducted a global spatiotemporal evaluation of ZTD data from TUW-VMF3 and GFZ-VMF3 using a 2020 ZTD dataset collected from 12,552 GNSS stations by the Karlsruhe Institute of Technology (KIT), analyzing their performance across regions, seasons, and times, and examining the influence of interpolation methods. The VMF data server offered two types of VMF ZTD products, namely VMF3_FC and VMF3_OP. VMF3_FC was released at 09:00 UTC the previous day for real-time use, while VMF3_OP was released at 18:00 UTC the next day (Boehm et al., 2006). Zhang et al. (2024) assessed the accuracy of these products using ZTD data from 255 CMONOC stations and found VMF3_OP to outperform VMF3_FC, especially in southeastern China. Yao et al. (2024) further analyzed both empirical and discrete VMF models using IGS ZTD products as benchmarks. Their findings indicated that site-specific models were preferred

when available, followed by network models. When discrete model data files were unavailable or when ZTD precision requirements were low, the GPT3 model was recommended.

In ZTD-related research, the processing of historical data has been well-established, with both ZTD models and associated products offering reliable data support. However, when the focus shifts to real-time data scenarios, the existing ZTD products exhibit significant limitations in real-time data generation capabilities, falling short of meeting the requirements for timeliness. Although various ZTD models possess the capability to predict ZTD values at any given moment, their forecast accuracy is generally low and insufficient for the needs of specific high-precision applications. It is worth noting that Numerical Weather Prediction (NWP) data offers the advantage of providing stratified atmospheric information, which can effectively support the processing and analysis of forecast data. From the perspective of ZTD calculation logic, setting the initial calculation value is essentially a forecasting process, and thus the availability and accuracy of predicted ZTD values play a decisive role in the reliability of research outcomes.

In recent years, researchers have proposed various innovative methods for tropospheric delay estimation using NWP systems. Zhang et al. (2022) combined GNSS-ZTD observations with NCEP-GFS forecasts, taking advantage of the good spatial coverage of NWP data and the high accuracy of GNSS, to develop the real-time wide-area precise tropospheric correction system (WAPTCs) for China. Yang et al. (2025) combined NCEP-GFS ensemble forecasts with GNSS observations and used XGBoost to calibrate the NCEP-GFS ZTD residuals based on GNSS, achieving ZTD predictions. Li et al. (2025) developed the HDTM model based on NCEP-GFS and ECMWF-IFS, which provides hourly updated grids of ZHD and weighted mean temperature (T_m), enabling accurate GNSS PWV retrieval. In addition to traditional NWP methods, artificial intelligence (AI) has been increasingly applied to atmospheric parameter forecasting. Wuhan University developed TropNet, which integrated GOES-R satellite observations with NCEP GFS forecasts, achieving ZWD predictions at 2 km spatial resolution and 1-hour temporal resolution (Lu et al., 2023). The FuXi model from Shanghai AI Laboratory, based on ERA5 reanalysis data, provided global forecasts for up to 15 days (Chen et al., 2023). The FengWu-GHR, developed by the Shanghai AI Laboratory, was the first AI-based system to model and forecast medium-range weather at a 10-kilometer resolution (Han et al., 2024). Huawei Cloud introduced Pangu-Weather, which employs a 3D Earth-Specific Transformer (3DEST) with a hierarchical time-aggregation strategy, improving forecast accuracy while increasing computational efficiency by 10,000 times (Bi et al., 2023). At present, many studies have demonstrated the capability of AI-based weather models in forecasting atmospheric parameters. Huang et al. used the Pangu-Weather model

to predict T_s data (Pangu- T_s) and developed a T_m forecast model (RF- T_m) for China (RF- T_m), facilitating real-time retrieval of accurate PWV (Huang et al., 2025).

Overall, both traditional NWP systems and AI-based weather models offer the potential for high-resolution, real-time ZTD forecasts. However, AI-based weather models mostly rely on reanalysis data, producing outputs for surface and atmospheric variables rather than ZTD directly. Moreover, the large computational demands and extensive data requirements limit their widespread adoption in typical research settings. In contrast, systematic research that integrates NWP data into ZTD computation and accuracy evaluation remains limited. Nevertheless, deeply incorporating NWP data into ZTD estimation has strong potential to enhance forecast reliability and accuracy, while providing more precise technical support for navigation and positioning applications, offering significant scientific value and practical prospects.

This study proposed a method for predicting ZTD using meteorological ensemble forecast data and analyzed the performances of the ensemble forecast data from different NWP systems. The ZTD observations from IGS stations were used as references, and the ensemble forecast data from two high-resolution NWP systems, ECMWF-IFS and NCEP-GFS, were selected to conduct experiments to analyze their specific performance in ZTD prediction. The article was structured as follows: Section 2 introduced the two types of ensemble forecast datasets and the methods used for ZTD calculation. Section 3 described the reference data and experimental methodology. Section 4 presented the analysis and discussion, and Section 5 concluded the study.

2. Data and methodology

2.1. Ensemble forecast data

The IFS developed by the ECMWF is one of the most advanced global numerical weather prediction systems. As part of its Open Data initiative, ECMWF provides public access to a subset of IFS forecast data free of charge. The ECMWF-IFS generates forecasts four times daily at 00:00, 06:00, 12:00, and 18:00 UTC. Forecasts initialized at 00:00 and 12:00 UTC extend up to 360 h into the future, with a temporal resolution of 3 h for the first 144 h and 6 h from 150 to 360 h. Forecasts initialized at 06:00 and 18:00 UTC extend up to 144 h into the future with a 6-hour temporal resolution. The spatial resolution is 0.25° for all four initial forecast times. The ECMWF-IFS forecast data includes 13 pressure levels of meteorological parameters, such as temperature, humidity, wind speed, and precipitation, provided in the GRIB2 format. These datasets are widely used in weather forecasting, climate research, disaster monitoring, and environmental protection, and they play a critical role in enhancing the accuracy of global meteorological information.

The GFS developed by the NCEP in the United States, is a key global numerical weather prediction system primarily designed for short-term forecasting. The NCEP-GFS provides global forecasts of surface and upper-atmospheric meteorological variables. Forecasts are initialized four times daily at 00:00, 06:00, 12:00, and 18:00 UTC and extend up to 384 h ahead. The forecast has a temporal resolution of 1 h for the first 120 h, and 3 h from 120 to 384 h. The spatial resolution is consistently 0.25°. NCEP-GFS forecast data includes 41 pressure levels of key meteorological variables such as temperature, humidity, wind speed, and precipitation, and are widely used in short-term weather forecasting, climate monitoring, and disaster management. NCEP-GFS data are distributed in GRIB2 format and have applications across various sectors, including aviation, maritime navigation, and environmental monitoring.

2.2. ZTD calculation using ensemble forecast data

The layered meteorological data, including parameters such as geopotential height, specific humidity, temperature, and pressure, are collected from ensemble forecast outputs to predict ZHD and ZWD separately. For ZHD calculation, a hybrid approach combining empirical modeling and numerical integration is employed. Considering the vertical gradient characteristics of atmospheric refractivity, ZHD estimation relies on atmospheric parameters such as pressure, temperature, and water vapor pressure. The study incorporates the Saastamoinen correction model (Saastamoinen, 1972) to establish a layered computation framework based on pressure-level data. First, the hydrostatic delay for each layer is calculated using the corresponding meteorological variables. Then, the trapezoidal rule is applied to numerically integrate these layers and obtain discrete ZHD values. Finally, an empirical correction is applied to account for the delay contribution from the top of the atmosphere, based on the station's latitude and the height of the uppermost layer. The specific steps and formulas are as follows:

$$N_{\text{dry}} = 77.689 \times \frac{P - e_s}{T} \quad (1)$$

where P represents air pressure, T represents temperature, and e_s represents vapor pressure. Thus, the refractivity delay of dry air, namely N_{dry} , is computed based on pressure-level data.

Between each atmospheric pressure layer, the trapezoidal rule is used to integrate to calculate the dispersive part of ZHD:

$$ZHD_{\text{discrete}} = \sum_{k=1}^{n-1} \frac{1}{2} (N_{\text{dry},k} + N_{\text{dry},k+1}) \times (\text{height}_{k+1} - \text{height}_k) \quad (2)$$

where n is the number of atmospheric pressure layers, and height_k is the height of the k -th layer.

To account for the delay from the pressure top layer to atmospheric pressure, the Saastamoinen model is used for correction (Shi et al., 2015; Yang et al., 2023b).

$$ZHD_{\text{top}} = 0.0022768 \times \frac{P_{\text{top}}}{(1 - 0.00266 \cos 2\varphi - 0.00028 \text{height}_{\text{top}})} \quad (3)$$

where φ represents the latitude of the station, and $\text{height}_{\text{top}}$ represents the top layer height.

The final ZHD is:

$$ZHD = ZHD_{\text{discrete}} \times 10^{-6} + ZHD_{\text{top}} \quad (4)$$

Then, a layered integration framework consistent with that used for ZHD is adopted to compute the ZWD (Yang et al., 2021c). Specifically, to address the pronounced variability of the wet component, an integral expression is formulated based on the vertical distribution of specific humidity, and discrete ZWD values are derived through numerical integration. The corresponding computational expressions used in the process are as follows:

$$N_{\text{wet}} = 71.2952 \times \frac{e_s}{T} + 375463 \times \frac{e_s}{T^2} \quad (5)$$

$$ZWD_{\text{discrete}} = \sum_{k=1}^{n-1} \frac{1}{2} (N_{\text{wet},k} + N_{\text{wet},k+1}) \times (\text{height}_{k+1} - \text{height}_k) \quad (6)$$

$$ZWD = ZWD_{\text{discrete}} \times 10^{-6} \quad (7)$$

Based on the defined computational methodology, the ZHD and ZWD values at each target station are calculated using the two forecast datasets. The final ZTD value is then obtained by summing the ZHD and ZWD components, as shown below:

$$ZTD = ZHD + ZWD \quad (8)$$

2.3. Unified heights and vertical interpolation

To ensure consistency between the IGS reference data and the NWP systems outputs, it is necessary to unify the height reference systems, convert the ensemble forecast data to a common height framework, and then apply linear interpolation or extrapolation to obtain meteorological variables at the target IGS station height. The detailed procedures are as follows:

In this study, the IGS station data provide ellipsoidal heights, whereas the ensemble forecast data are referenced to geopotential heights. Since the two height systems are not directly comparable, a unification of the height reference is required. The conversion from geopotential height to geometric height is performed using the following relation:

$$h = \frac{RH}{R - \frac{g(\varphi)}{g_0} H} \quad (9)$$

$$H = \Phi/g_0 \quad (10)$$

where h represents the geometric height, H represents the geopotential height, Φ is the geopotential, R is the mean radius of the Earth, $g(\varphi)$ is the gravitational acceleration at latitude φ , and $g_0 = 9.80665 \text{ m/s}^2$ is the standard gravity value.

After converting the vertical levels of the ensemble forecast data into geometric heights, including pressure, temperature, and specific humidity are obtained through vertical interpolation or extrapolation, depending on the relative position of the station height to the NWP vertical layers.

When the IGS station height lies within the range of the ensemble forecast data layers, a layer-wise linear interpolation scheme is applied to derive the meteorological variables at the station height. This approach enables a continuous representation of the vertical variation of atmospheric parameters within the layers, ensuring both smoothness and physical consistency in the interpolation process.

$$X(h_s) = X(h_{k-1}) + \frac{X(h_k) - X(h_{k-1})}{h_k - h_{k-1}}(h_s - h_{k-1}) \quad (11)$$

where X represents any atmospheric variable, including pressure P , temperature T , and specific humidity q , and $X(h_{k-1})$ and $X(h_k)$ are the values of variable X at those two layers.

When the IGS station height is below the lowest pressure level of the ensemble forecast data, vertical extrapolation is required. For temperature and specific humidity, linear extrapolation is used.

$$T(h_s) = T_1 + \frac{T_2 - T_1}{h_2 - h_1}(h_s - h_1) \quad (12)$$

$$q(h_s) = q_1 + \frac{q_2 - q_1}{h_2 - h_1}(h_s - h_1) \quad (13)$$

where h_1 and h_2 represent the heights of the lowest two layers.

According to the hydrostatic equation, atmospheric pressure decreases exponentially with height; therefore, pressure is extrapolated using an exponential decay formulation:

$$P(h_s) = P_1 \exp[-e_c(h_1 - h_s)] \quad (14)$$

where P_1 is the pressure at the lowest NWP level and e_c is the local lapse-rate-related coefficient.

3. Experiment

3.1. IGS-ZTD reference data

Since 1997, the international GNSS services (IGS) has produced ZTD products through the systematic integration of multi-center ZTD estimates derived from its participating analysis centers (Dow et al., 2009; Flores et al., 2000). The current IGS-ZTD product suite encompasses all operational IGS stations archived in official data repositories, boasting a stated accuracy of 1.5–5 mm, though

empirical analyses suggested that actual errors tend to be marginally larger (Byun et al., 2009).

In this study, the ZTD product released by IGS with a temporal resolution of 5 min was utilized as reference data. Data availability was evaluated for 462 IGS stations around the world from March 17 to April 16, 2025, of which 12 stations had less than 10 days of available data and were excluded from subsequent analyses. The distribution of the selected IGS stations was illustrated in Fig. 1.

3.2. Statistical metrics

To assess the accuracy of the ZTD forecast values calculated from the ECMWF-IFS and NCEP-GFS forecast data, the root mean square error (RMSE) and bias were used as evaluation metrics in this study. RMSE is a common method to quantify the difference between forecast and actual values. Bias indicates the difference between the expected value predicted by the model and the true value, indicating the overall deviation of the model in an average sense. The specific calculation formulas are as follows:

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (ZTD_i - ZTD_i^R) \quad (15)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (ZTD_i - ZTD_i^R)^2} \quad (16)$$

where ZTD_i is the estimated value of the tropospheric delay, ZTD_i^R is the reference value of the tropospheric delay, i and N respectively represent the i value and the total number of samples.

3.3. Interpolation methods

In this study, since the two ensemble forecast data are computed in 3-hour time steps, the ZTD reference data provided by IGS stations are extracted in 1-hour time steps. To ensure that the time resolution of the experimental and reference datasets is consistent for accuracy assessment, the ZTD forecasts calculated from the ensemble forecast data are linearly interpolated to match the reference ZTD data, which is shown as follows:

$$y = y_0 + \left(\frac{y_1 - y_0}{x_1 - x_0} \right) * (x - x_0) \quad (17)$$

where x represents the independent variable value of the estimated point, and y is the corresponding estimated value of the dependent variable.

4. Analysis and discussion

To show the overall performance of two ensemble forecast data, their ZTD forecast values were illustrated as the density scatter plots using the IGS-ZTD as references in

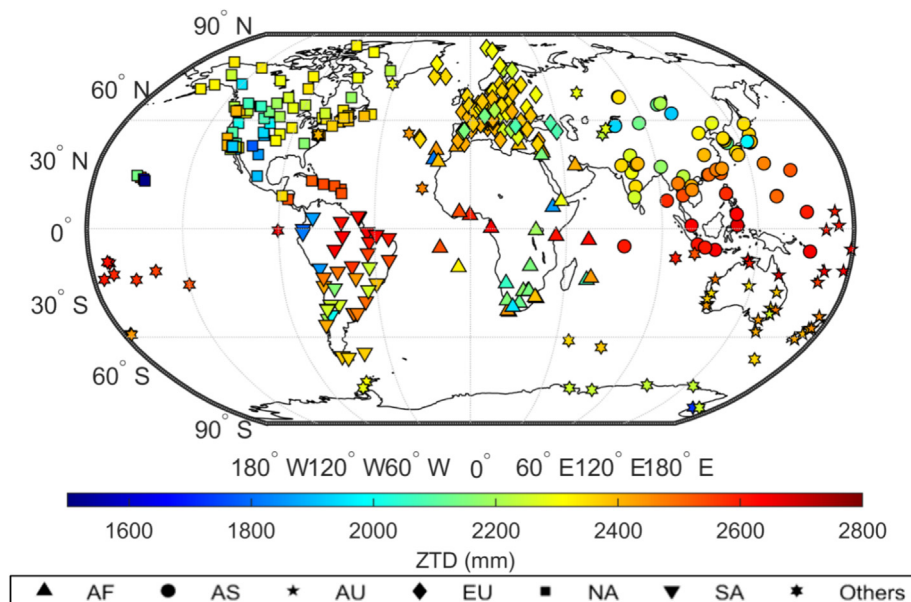


Fig. 1. Distribution map of IGS stations.

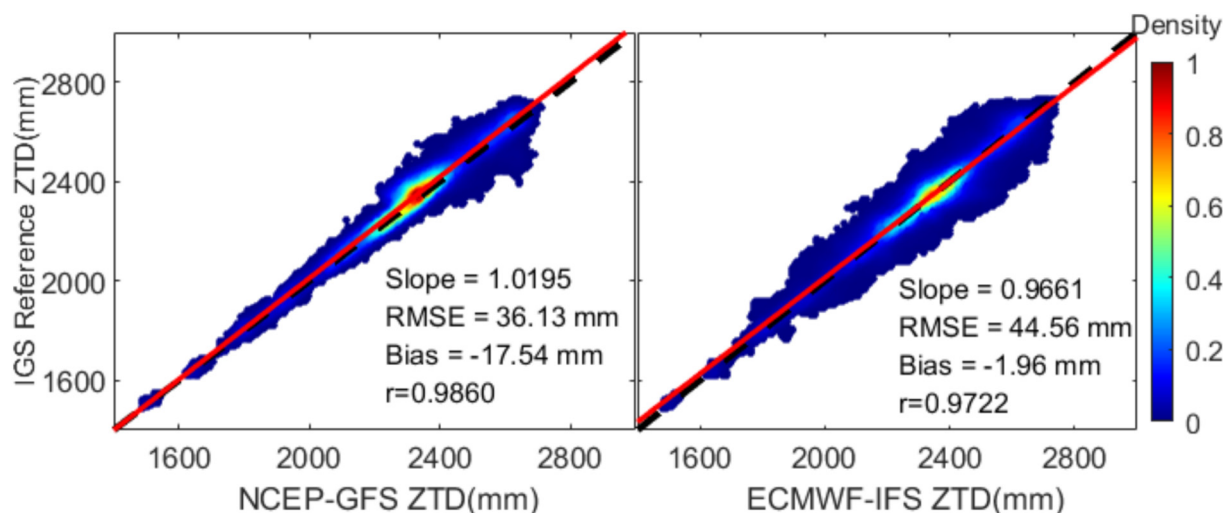


Fig. 2. Density scatter plots between the ensemble forecast ZTD and the reference ZTD.

Fig. 2. The black dashed line in the figure represented the 1:1 reference line, while the red solid line was the linear regression fit of the forecast data. The results showed that the ZTD predictions from both ECMWF-IFS and NCEP-GFS exhibit a good linear consistency with the IGS reference values, with correlation coefficients of 0.9722 and 0.9860, respectively.

From the left panel, the slope of the linear regression line was 1.0195 for the NCEP-GFS ZTD, which was slightly greater than 1, indicating a slight overall overestimation, particularly evident in regions with higher ZTD values. For the ECMWF-IFS ZTD, the regression slope was 0.9661, which was slightly less than 1, reflecting a minor underestimation overall. The bias and RMSE were $-17.54/36.13$ mm and $-1.96/44.56$ mm for the two

ensemble forecast data, respectively. The NCEP-GFS ZTD had better RMSE, and correlation compared to the ECMWF-IFS ZTD, while larger bias also existed in the NCEP-GFS ZTD.

To further analyze the applicability and error characteristics of ECMWF-IFS and NCEP-GFS ensemble forecast data in ZTD prediction, a residual analysis was conducted and shown in the form of a histogram in Fig. 3. The residuals of ECMWF-IFS ZTD exhibited a relatively wide distribution but were centered around 0, with a mean bias of -1.96 mm and a standard deviation (SD) of 44.52 mm. The residual distribution of NCEP-GFS ZTD was relatively more concentrated but showed a clear left-skewed pattern, with bias and SD of -17.54 mm and 31.59 mm, respectively. The percentage of ZTD residuals smaller than

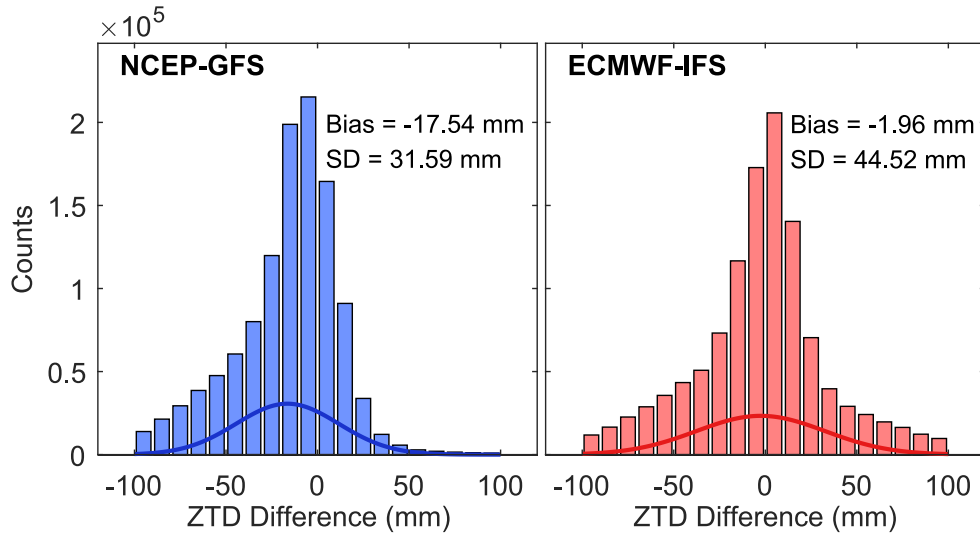


Fig. 3. Histogram of ZTD residual distribution between ensemble forecast data and reference values.

20 mm in both ensemble forecast data exceeded 70 %, and the percentage reached 90 % when the value was set to smaller than 50 mm.

Fig. 4 presented a comparison of the error characteristics of ZTD forecasts, which was made through density histograms of RMSE and bias. It not only illustrated the statistical distribution of errors across all stations, but also included thin lines representing the individual station error values at the bottom. In terms of RMSE distribution, both ensemble forecast data exhibited relatively low RMSE values, with the mean centered around 15 mm. However, the lower scatter plot showed that the ECMWF-IFS ZTD had a significantly larger number of high error stations than the NCEP-GFS ZTD when the RMSE exceeded 30 mm, suggesting that the overall stability of the ECMWF-IFS ZTD was slightly lower. In terms of the bias distribution, the bias values of both ensemble forecast data were close

to zero with the main peak of the ECMWF-IFS ZTD around 10 mm, while the main peak of the NCEP-GFS ZTD around -10 mm indicating systematic underestimation. The deviation scatter plots further showed that the deviations of both NCEP-GFS ZTD and ECMWF-IFS ZTD were mainly concentrated between -40 mm and 40 mm. The distributions were more symmetrical and balanced, indicating comparable ZTD forecast performance.

To highlight the differences of the forecast data across various stations, Table 1 summarizes the RMSE and Bias of the two ZTD datasets calculated for each IGS station, including the maximum, minimum, and mean values. For ECMWF-IFS ZTD, from 8.12 mm to 83.30 mm, with a mean value of 38.57 mm. For NCEP-GFS ZTD, the RMSE ranged from 8.11 mm to 77.63 mm, with a mean value of 30.66 mm. It indicated that although NCEP-GFS ZTD had a better average value, its RMSE range at

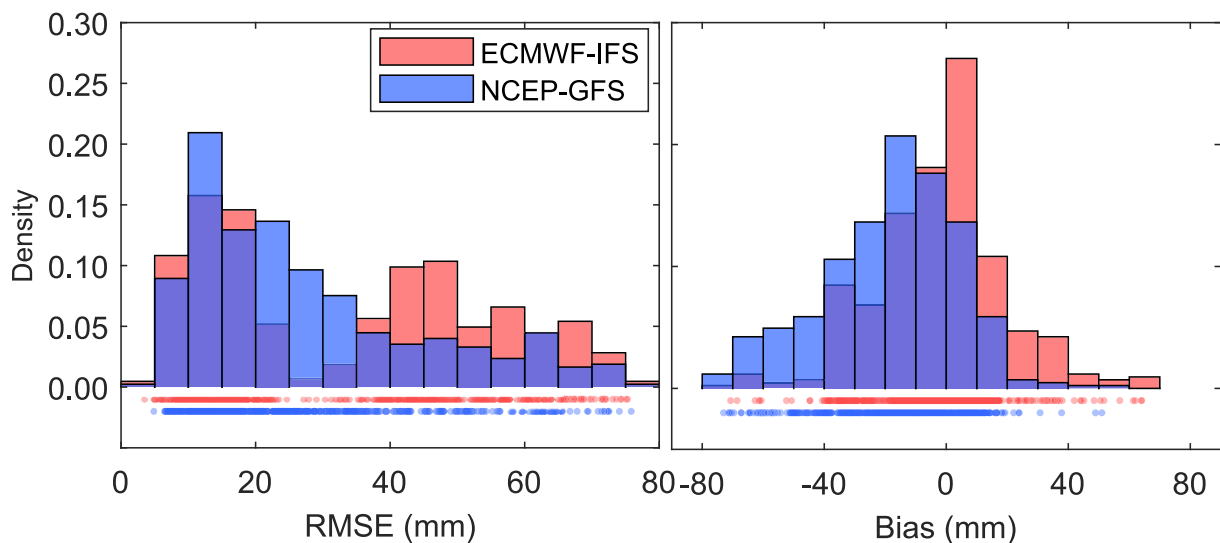


Fig. 4. Histogram of RMSE and bias density distributions for the two types of ensemble forecast data.

the IGS stations was larger and the station with the highest RMSE value also appeared in this type of ensemble forecast data. In terms of bias, NCEP-GFS ZTD showed a more notable negative mean bias, with a larger number of stations clustered in the negative-bias range. In contrast, the bias distribution of ECMWF-IFS ZTD was more symmetric, with a mean value of -2.18 mm. This indicated that NCEP-GFS demonstrates higher error variability across space, with larger fluctuations among different regions. Overall, ECMWF-IFS performed better in terms of systematic bias and was more suitable for applications requiring high consistency and overall accuracy, while NCEP-GFS was more sensitive to regional atmospheric differences, and this higher variability made it more appropriate for multi-station or large-area studies.

To illustrate the spatial distribution of RMSE and bias for each station, the corresponding results of the two types of ensemble forecast data were shown in Fig. 5. The two types of ensemble data performed well with most RMSE values below 35 mm and bias ranged between ± 60 mm. Notably, the ECMWF-IFS ZTD exhibited clear regional differences, with elevated RMSE values particularly evident

in parts of Asia, Africa, and Oceania—some exceeding 70 mm and reaching up to 89 mm. In contrast, RMSE values in North and South America were significantly lower, with several stations approaching 0 mm. This pattern was not observed in the NCEP-GFS ZTD, which demonstrated a more uniform RMSE distribution across regions. As for bias, the spatial distribution was broadly similar between the two ensemble forecast data. Negative biases were predominantly found in South America and Europe, whereas positive biases were more common in North America and Asia. The regional differences between ECMWF-IFS and NCEP-GFS are influenced not only by meteorological regimes but also by differences in the vertical pressure-level structures and physical parameterizations of the two forecast systems. In regions such as the Asian monsoon area, where vertical humidity gradients are strong, the limited number of IFS vertical layers reduces its ability to accurately represent the humidity profile, thereby amplifying ZWD-related ZTD errors. In addition, many high-error stations are in mountainous or strongly convective regions, where intrinsic atmospheric variability further increases uncertainty.

To analyze their performance at different latitudes and elevations, the stations were grouped into four elevation categories and six latitude zones. Four elevation categories were defined as E1 (0–60 m, 110 stations), E2 (61–140 m, 117 stations), E3 (141–560 m, 111 stations), and E4 (above 560 m, 112 stations). The stations were grouped into six latitude zones with 30° intervals, including SL1 (90°S – 60°S), SL2 (60°S – 30°S), SL3 (30°S – 0°), NL1 (0° – 30°N), NL2 (30°N – 60°N), and NL3 (60°N – 90°N), as shown in Fig. 6.

Table 1
Table of data error statistics.

	NCEP-GFS		ECMWF-IFS	
	RMSE (mm)	Bias (mm)	RMSE (mm)	Bias (mm)
Max	77.63	51.51	83.30	63.71
Min	8.11	−74.18	8.12	−70.79
Mean	30.66	−17.71	38.57	−2.18

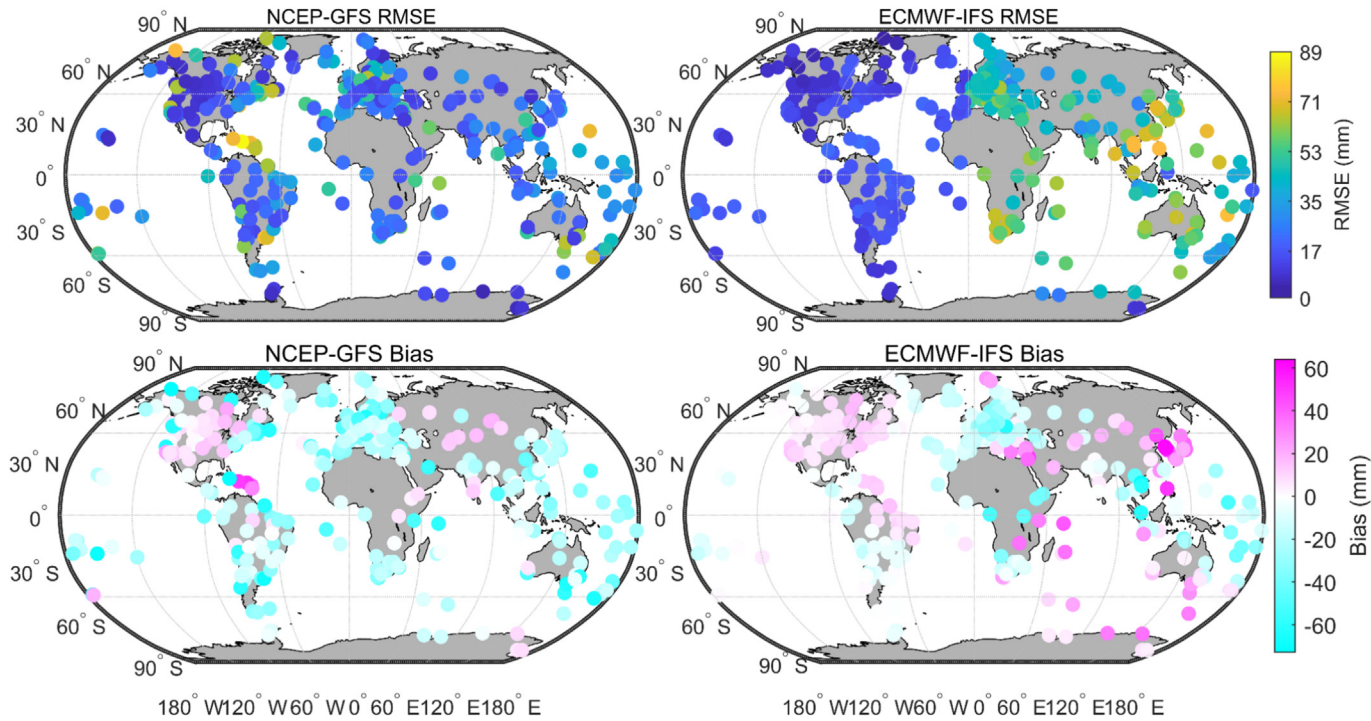


Fig. 5. Global error distributions for the two types of ensemble forecast data.

From Fig. 6, the ECMWF-IFS ZTD in the E1 category exhibited an RMSE and bias of 26.67/4.32 mm, indicating lower error and more stable performance compared to the NCEP-GFS ZTD at 48.80/−36.58 mm. In contrast, the NCEP-GFS ZTD demonstrated better performance in the E3 and E4 categories. ECMWF-IFS ZTD showed lower RMSE and more stable bias at low elevations, and its overall accuracy surpassed that of NCEP-GFS ZTD. However, at higher elevations, NCEP-GFS ZTD provided relatively better forecast accuracy. Regarding latitude, ECMWF-IFS ZTD showed stronger consistency and stability in the middle and high latitude regions, with smaller bias and more balanced forecasts, while NCEP-GFS ZTD had some advantages in the high latitude regions of the Southern Hemisphere, but there was a systematic underestimation in the high latitude regions of the Northern Hemisphere. For example, the RMSE and bias of ECMWF-IFS ZTD and NCEP-GFS ZTD in SL1 were 19.66/8.42 mm and 9.40/−0.20 mm, respectively, reflecting the highest accuracy among all latitude zones and suggesting enhanced stability in high-latitude regions. Across SL2, SL3, NL1, and NL2, both datasets maintained RMSE values between 0 mm to 50 mm, with ECMWF-IFS generally showing smaller biases than NCEP-GFS. ECMWF-IFS was more suitable for applications in low-elevation and mid-to-high latitude regions, whereas NCEP-GFS exhibited advantages in high-latitude and complex terrain areas of the southern hemisphere.

To conduct a more detailed analysis of the performance of the proposed method in specific regions, the IGS stations were categorized into seven groups, which were North America (NA, 109 stations), Europe (EU, 111 stations), Asia (AS, 62 stations), Australia (AU, 40 stations), South America (SA, 43 stations), Africa (AF, 37 stations),

and Other regions (Others, 48 stations). The classification of the IGS stations can be well seen from Fig. 1.

The density scatter plots were utilized for the two ensemble forecast data in these seven regions, which illustrated their predictive capabilities and regional differences in ZTD prediction (see Fig. 7). Overall, the slopes of the NCEP-GFS ZTD were slightly larger than 1 in most regions, indicating a slight overestimation tendency, while the slopes of the ECMWF-IFS ZTD were mostly smaller than 1, showing some underestimation tendency. Despite these systematic differences, both ensemble forecast data maintained strong correlations with the IGS ZTD observations, demonstrating their overall reliability in ZTD prediction. In AS, the slope of NCEP-GFS ZTD was 1.0290, with RMSE and bias of 31.06/−15.78 mm; while the slope of ECMWF-IFS ZTD was 0.9485, with RMSE and bias of 67.25/10.19 mm. NCEP-GFS ZTD achieved its lowest RMSE among all regions in this area, indicating its superior performance in regions influenced by monsoons and strong convective activity. In EU, the slope of ECMWF-IFS ZTD dropped to 0.8368, with RMSE and bias of 48.25/−18.66 mm, showing a significant underestimation. In contrast, NCEP-GFS ZTD yielded an RMSE of 33.90 mm in this region, exhibiting overall better performance. In SA, although the slope of the ECMWF-IFS ZTD was slightly lower, its RMSE was only 16.98 mm, which was the lowest among all regions, and its bias was closest to zero. These results highlighted its particularly strong performance in tropical humid areas. Both ensemble forecast data demonstrated overall stable performance, with similar error levels and slight deviations in some regions, indicating that they have their own adaptability and application potential in different climatic environments.

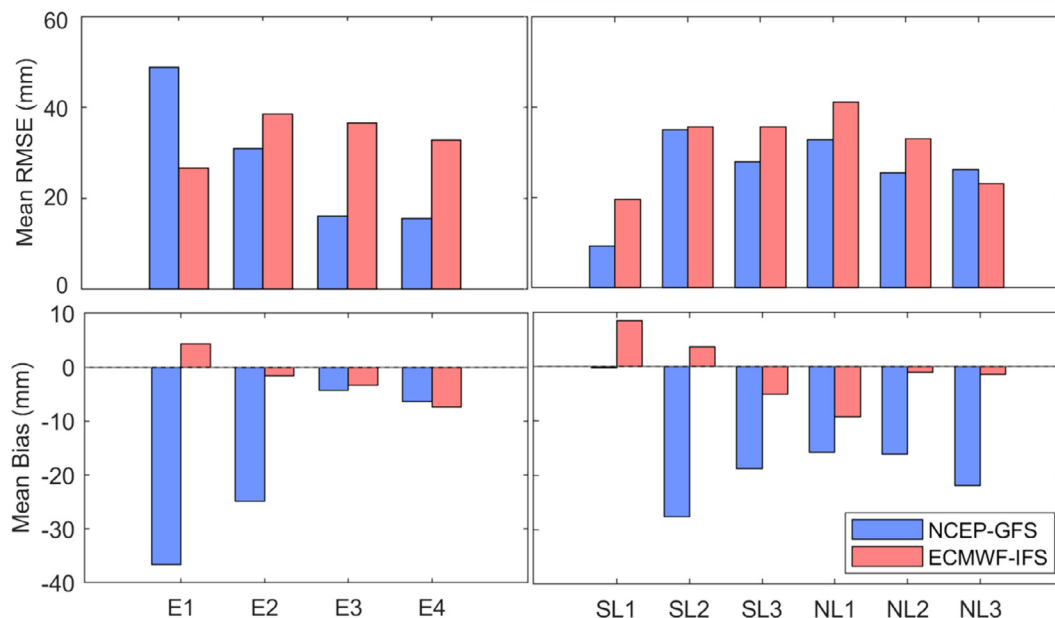


Fig. 6. Mean RMSE and bias across different latitude and elevation categories.

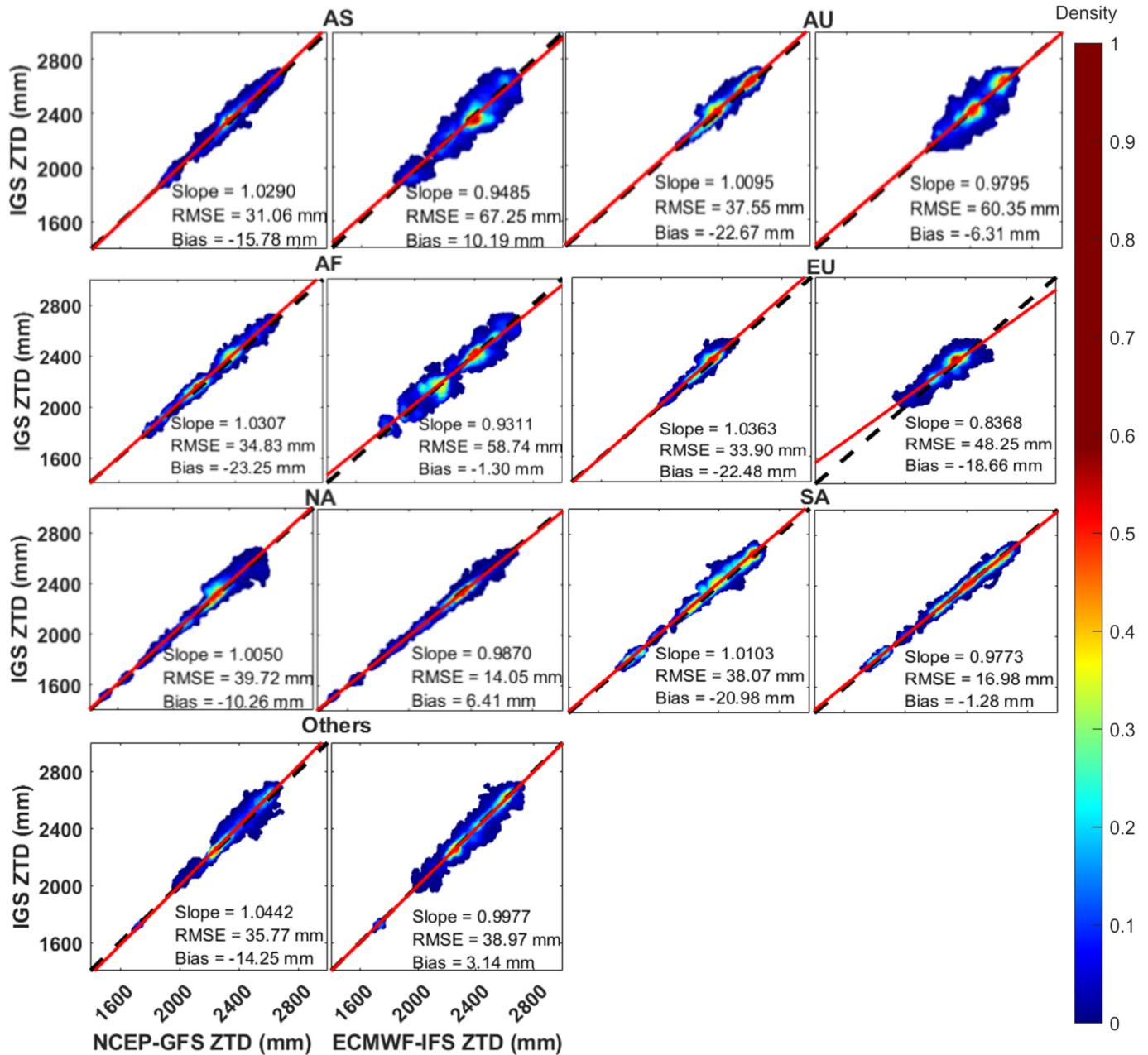


Fig. 7. Scatter density plot of correlation for different regions.

The violin plots in Fig. 8 showed the distributions of RMSE and bias for the two ensemble forecast data in each specific region. The first quartile (Q1) and the third quartile (Q3) indicated the rate of spread of the data, while the median (Q2) indicates the central tendency. The left panel showed that the RMSE of the two ensemble forecast data was mostly concentrated between 0 mm and 50 mm in the seven regions. The right panel showed that the median bias of the ECMWF-IFS ZTD was close to zero in most regions and was mostly concentrated between -15 mm and 15 mm. In contrast, the bias distribution of the NCEP-GFS ZTD was generally slightly wider, but the overall deviations remained relatively small. At low latitudes, the bias of

the ECMWF-IFS ZTD was concentrated between -40 mm and 40 mm, with some extreme values approaching ± 50 mm. Meanwhile, the NCEP-GFS ZTD showed a slight negative bias, mostly between -30 mm and 25 mm, indicating a tendency for localized underestimation in the tropics.

Since both types of ensemble forecast data provided four forecasts per day at 00:00, 06:00, 12:00, and 18:00 UTC, the ZTD forecast accuracy was analyzed for seven specific regions based on different initial forecast times. The results of this analysis were presented in the dual-axis plots shown in Fig. 9, where RMSE was illustrated using bar charts and bias was represented by lines. The

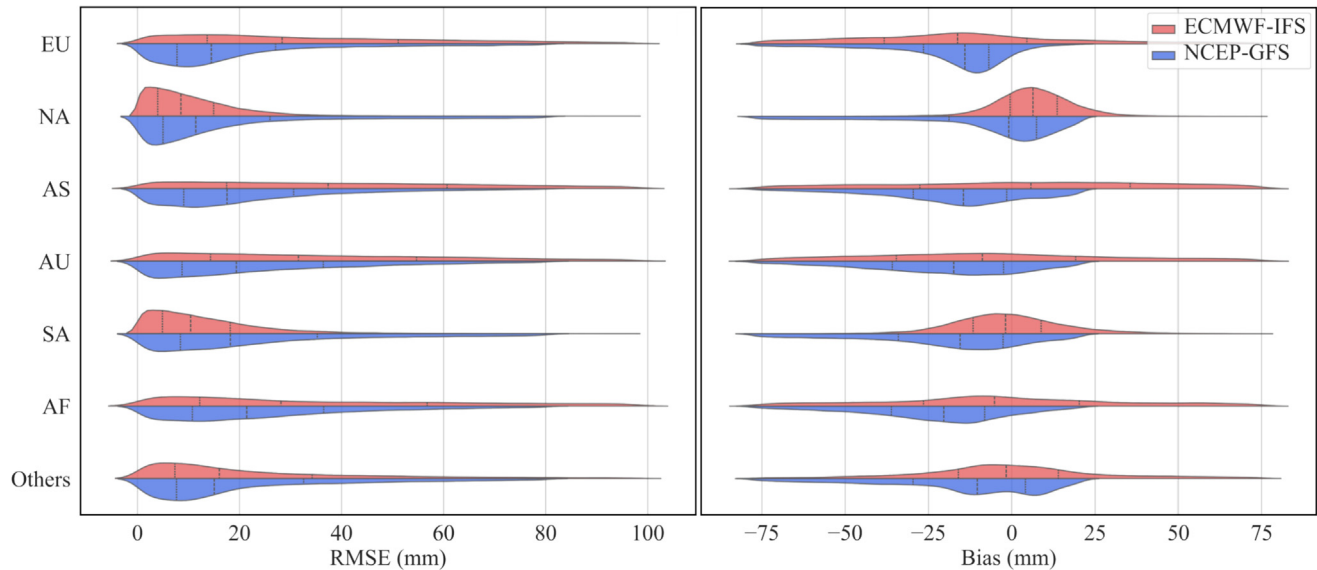


Fig. 8. Violin plots of ZTD forecast error distributions across seven regions for the two types of ensemble forecast data.

range between the maximum and minimum values was also marked in the figure. Overall, the RMSE of the two ensemble forecast datasets generally fell within the range of 0 mm to 70 mm, and the bias ranged from -60 mm to 80 mm. In most regions, such as EU, AF, AS and AU, the RMSE of ECMWF-IFS ZTD was generally higher than that of NCEP-GFS ZTD, whereas in NA and SA, the RMSE of ECMWF-IFS ZTD was lower. In other regions, the differences between the two types of ensemble forecast data in terms of RMSE and bias were smaller and showed higher consistency. The deviation curves and extreme value ranges

illustrated that the bias of ECMWF-IFS ZTD was closer to zero, but it was still found to be larger than NCEP-GFS ZTD in some regions. Note that significant negative bias could be observed in certain regions of the NCEP-GFS ZTD, suggesting that it may be more sensitive to complex terrain and climatic conditions.

In summary, the regional performance differences between the two types of ensemble forecast data resulted from the combined effects of regional meteorological characteristics, terrain conditions, and model-dependent representations of atmospheric processes. The interaction of

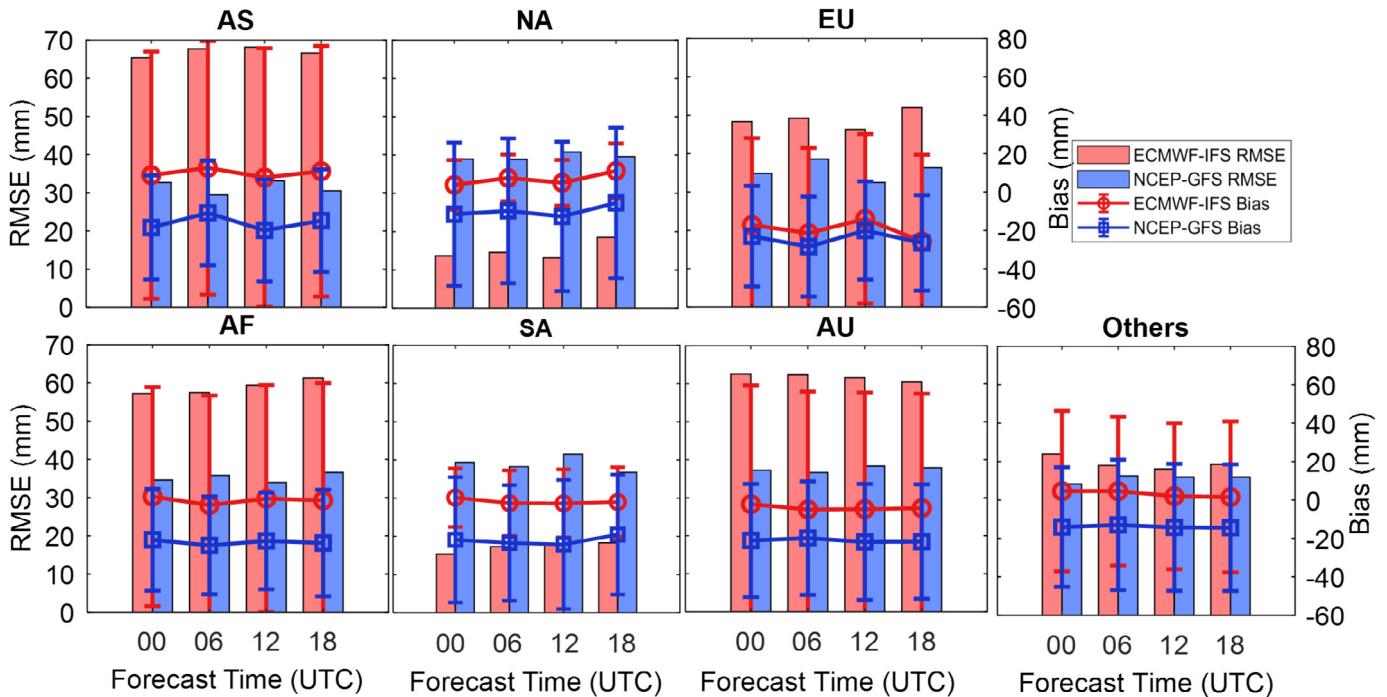


Fig. 9. Dual axis plots of RMSE and bias for different forecast initialization times for seven regions.

these factors led to the distinct spatial patterns observed in the ZTD prediction accuracy across seven specific regions.

To better understand the forecast performance of the ensemble forecast data, this study also examined the variability at different initialization times. Given that both types of ensemble forecast data provided continuous ZTD forecasts and their accuracy varied with initialization times, the temporal accuracy of the ZTD forecasts was systematically evaluated over a 360-hour forecast horizon. Specifically, box plots of RMSE and bias were used to illus-

trate how forecast errors changed over time under different initialization conditions. This approach helped clarify the performance trends throughout the entire forecast period.

Fig. 10 presented the ZTD time series and corresponding bias variations for seven representative IGS stations selected from different global regions. The stations were chosen based on the availability and completeness of ECMWF-IFS, NCEP-GFS, and IGS reference data during the study period, ensuring consistent comparison across datasets. As shown in the figures, the IGS ZTD remained

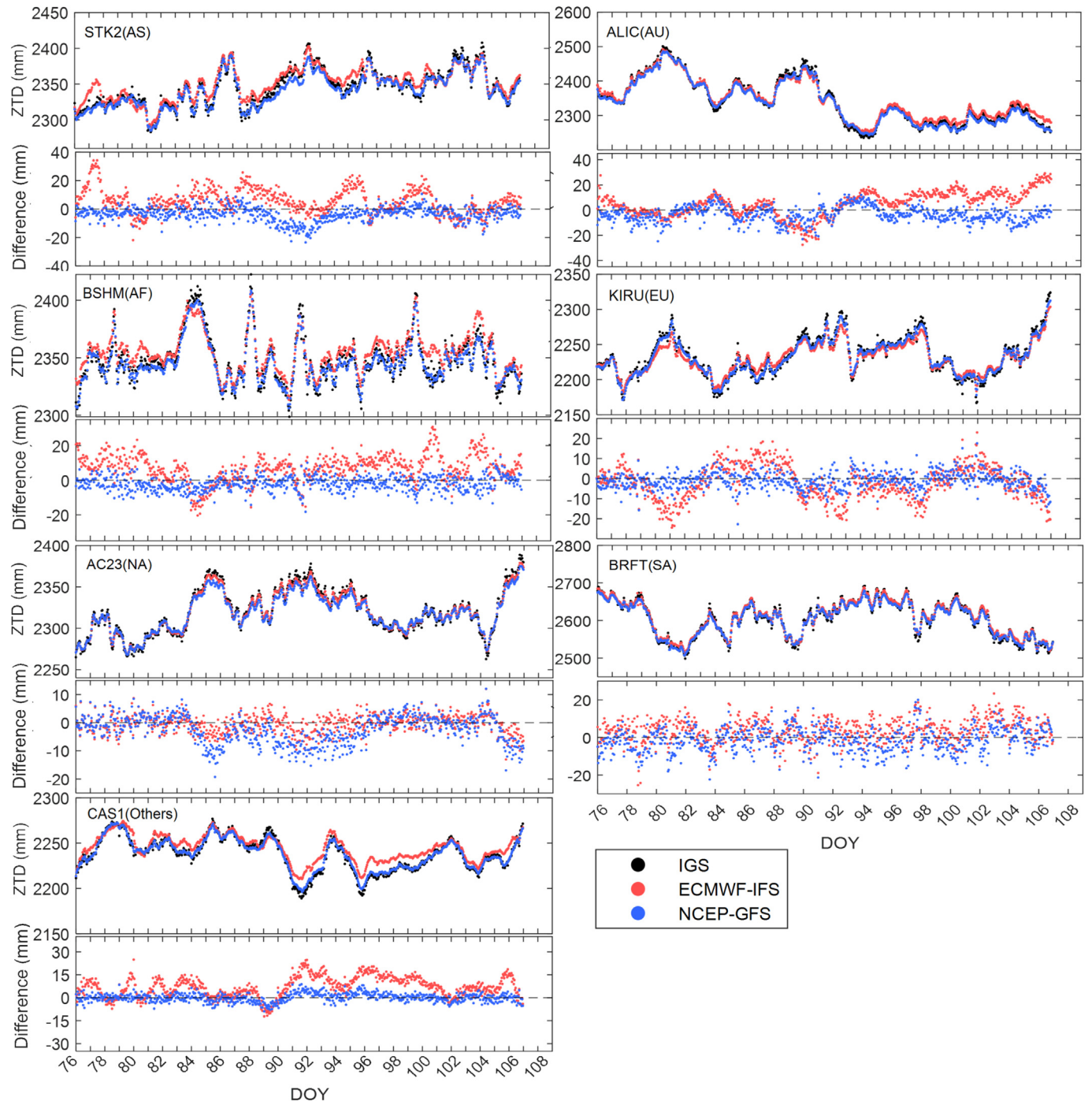


Fig. 10. Time-series analysis of ECMWF-IFS and NCEP-GFS forecasts and IGS ZTD reference values for seven representative global stations.

generally stable throughout the analysis period, exhibiting typical daily and short-term variations. Both ECMWF-IFS and NCEP-GFS successfully reproduced the overall fluctuation patterns observed in the IGS data, with comparable variation amplitudes, indicating good temporal stability in their forecasts. In terms of bias behavior, ECMWF-IFS showed slightly larger fluctuations at several stations, whereas NCEP-GFS exhibits a more stable and smoother bias evolution over time. Overall, the selected stations maintain bias values within approximately ± 40 mm, and both forecasting systems demonstrate reasonable capability in capturing the daily variability and regional characteristics of ZTD.

As shown in Fig. 11, the overall variability of the errors in the two types of ensemble forecast data remained small over time. The RMSE was controlled in the range of 0 mm to 60 mm, indicating high forecast stability and reliability across all initialization times. At 00:00 UTC and 12:00 UTC, the median RMSE of the ECMWF-IFS ZTD was maintained at about 25 mm, and the error distribution was concentrated, which indicated that its forecasting performance was stable. Although the RMSE of the forecast initialized at 18:00 UTC increased slightly, it was still comparable to that at 12:00 UTC. The overall performance of the NCEP-GFS ZTD was like that of the ECMWF-IFS ZTD, especially in the first five days, but with an increase in the RMSE in longer forecast periods. In terms of bias, the fluctuations of the two types of ensemble forecast data

with time were small, and most of the bias were between -30 mm and 30 mm with the median value closed to zero.

Furthermore, the representative stations with the best and worst forecast performance were selected at different initialization times, and their error distributions were depicted in Fig. 12, where the rectangles with diagonal texture represented the stations with the worst forecasting ability, and the solid-colored rectangles represented the best-performing stations. The upper panel showed that the RMSE of both ensemble forecast data at different initialization times was mainly within the range of 0 mm to 15 mm at the best stations, such as MONP and COTE. The RMSE of ECMWF-IFS ZTD exceeded 90 mm at PERT. Other poorly performing stations include NNOR and BTNG, where the RMSE of ECMWF-IFS ZTD generally exceeded 60 mm. For NCEP-GFS ZTD, a notably high RMSE, close to or above 80 mm, was observed at the SEYG. The lower panel showed that the bias of both ensemble forecast data was close to zero at the best stations. However, at stations with significant errors, distinct differences emerged between the two data. For instance, at PERT, the bias of ECMWF-IFS ZTD reached approximately 95 mm at 06:00 UTC, while NCEP-GFS showed no significant bias at this station. In contrast, NCEP-GFS ZTD showed a consistent underestimation at SEYG across all initialization times, with a maximum negative bias of approximately -80 mm. The WIND station also exhibited a negative bias of about -70 mm at 00:00 and

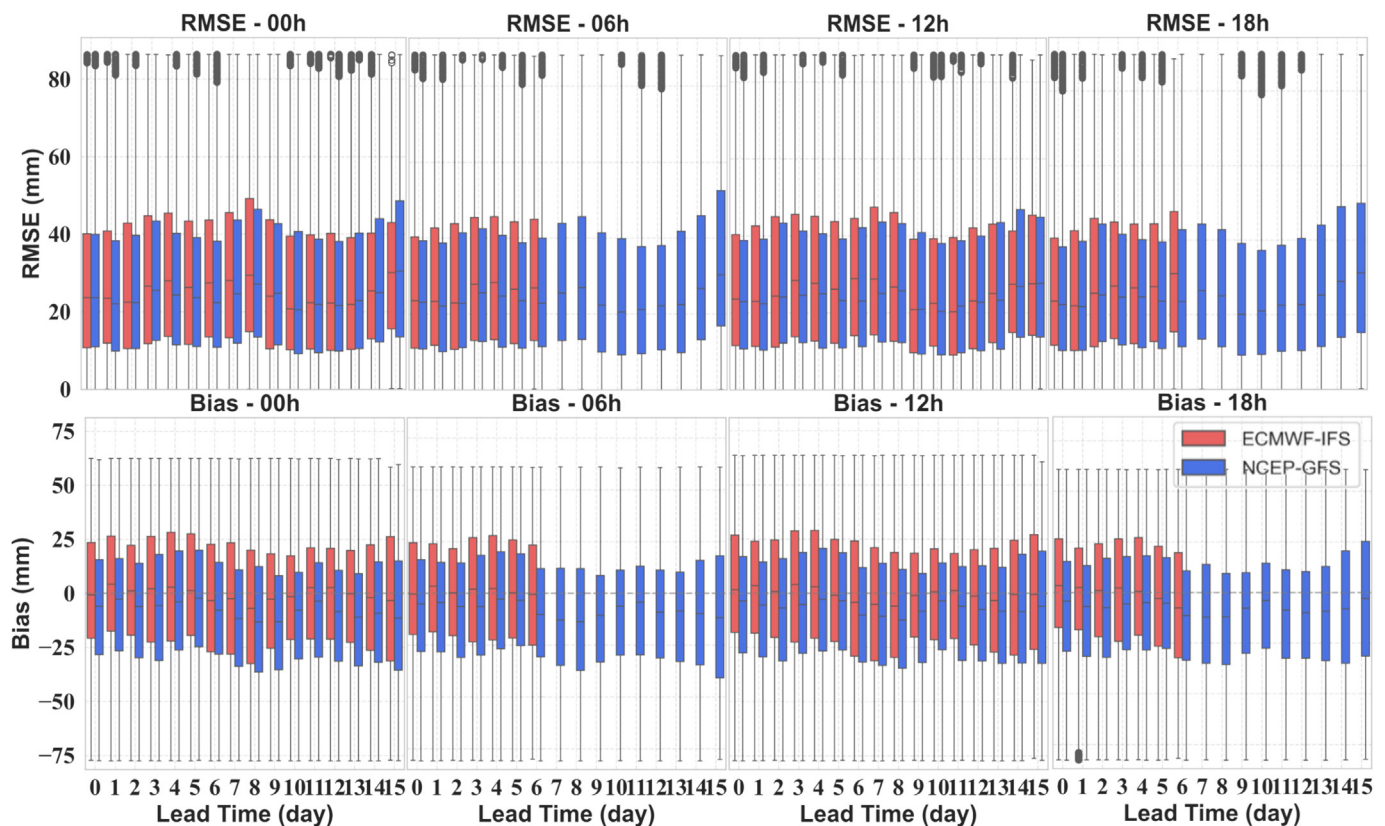


Fig. 11. Box plots of RMSE and bias of ZTD forecasts from ensemble forecast data over the forecast lead time.

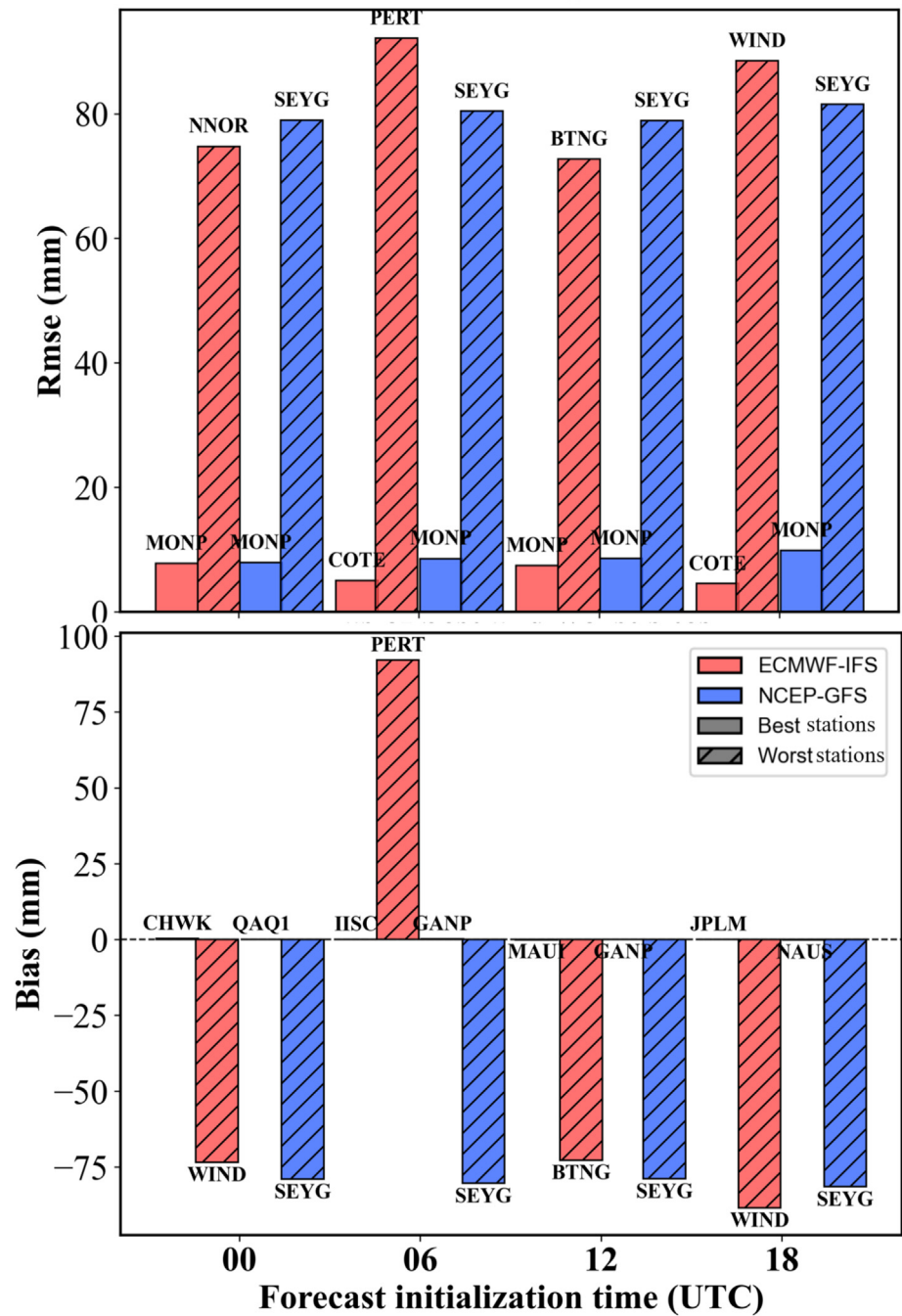


Fig. 12. Error analysis of the best and worst stations at different forecast initialization times.

18:00 UTC, which was mainly reflected in the ECMWF-IFS ZTD. High errors mainly stem from complex climatic conditions at the station locations. In equatorial humid regions (SEYG, BTNG), strong convection and high humidity cause rapid water vapor changes, amplifying ZTD forecast errors. In coastal regions (PERT, NNOR), frequent land–sea interactions and rapid pressure and temperature changes reduce forecast accuracy. Differences in vertical pressure levels between the two ensemble systems can also introduce ZTD discrepancies. In inland plateaus (WIND), high altitude, low pressure, and complex terrain make precise ZTD forecasts challenging.

5. Conclusion

This study proposed a method for predicting ZTD using meteorological ensemble forecast data and conducted the first systematic analysis of ZTD forecasting performance using two numerical weather prediction data sources, namely ECMWF-IFS and NCEP-GFS. A total of 450 IGS stations worldwide were selected as reference stations to evaluate the forecast accuracy of ECMWF-IFS ZTD and NCEP-GFS ZTD in terms of numerical prediction errors, spatial and temporal distribution characteristics, and the accuracy and reliability of real-time ZTD

forecasting and acquisition. Overall, both types of ensemble forecast data demonstrated good capability in predicting ZTD, although differences were observed across regions and forecast initialization times.

Specifically, ECMWF-IFS ZTD and NCEP-GFS ZTD exhibited strong linear consistency with the IGS reference values, with correlation coefficients of 0.9722 and 0.9860, respectively. The average RMSE and biases of the two types of ensemble forecast data were 38.57/–2.18 mm and 30.66/–17.71 mm. Further residual analysis showed that at 90 % of the stations, the residuals for both the ECMWF-IFS ZTD and the NCEP-GFS ZTD were clustered within ± 50 mm. NCEP-GFS was more suitable for applications requiring long-term stability and large-scale consistency, whereas ECMWF-IFS was better suited for regions demanding higher accuracy and sensitivity to systematic bias.

From a global spatial perspective, the two types of ensemble forecast data demonstrated good ZTD prediction capability at most stations, with RMSE values primarily concentrated between 0 and 35 mm. The forecasting performance of the two types of ensemble forecast data showed significant spatial variation with respect to elevation, latitude, and regional characteristics. ECMWF-IFS ZTD performed better in low-elevation areas, with an RMSE and bias of 26.67 mm and 4.32 mm, respectively, significantly outperforming NCEP-GFS ZTD, which had an RMSE and bias of 48.80 mm and –36.58 mm. Conversely, in mid-to-high elevation regions, NCEP-GFS ZTD provided more stable forecasts with smaller errors. Regarding different latitude zones, ECMWF-IFS ZTD exhibited smaller biases and more concentrated distributions in mid-to-high latitudes, while the NCEP-GFS ZTD had the smallest error in the high latitudes of the southern hemisphere, with an RMSE of 9.40 mm and a bias of only –0.20 mm, showing a stronger adaptation to the polar regions. In terms of performance across seven specific regions, ECMWF-IFS ZTD excelled in South America, achieving the lowest RMSE of 16.98 mm and near-zero bias. However, there was regional variability in Asia, Africa, and Australia, where forecast accuracy was relatively low. NCEP-GFS ZTD performed best in Asia, with an RMSE of 31.06 mm and a bias of –15.78 mm. In North America and Europe, NCEP-GFS ZTD tended to exhibit a negative bias.

From a temporal perspective, both types of ensemble forecast data exhibited overall stable ZTD prediction errors across different initialization times, with RMSE mostly controlled within 0 mm to 60 mm. ECMWF-IFS ZTD showed higher accuracy at 00:00 and 12:00 UTC, with median RMSE around 25 mm and concentrated error distributions, while NCEP-GFS ZTD errors slightly increased with forecast lead time. Specifically, at well-performing stations, the RMSE from both types of ensemble forecast data generally remained below 15 mm across all initialization times, with biases close to zero, demonstrating good temporal consistency in forecasts, whereas

poorly performing stations exhibited large errors. Stations affected by strong convection, coastal influences, or complex terrain still exhibit noticeable biases, indicating that achieving high-accuracy ZTD forecasts remains challenging in regions with highly complex meteorological conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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